

Design tools: drawing the solution before you build it

Four ways to put a design on paper — mock-ups, sitemaps, storyboards and pseudocode. Same solution, four different lenses. Knowing which lens shows what is the whole game in the exam.

What this key knowledge covers

This summary distils everything you need for **KK07 — Design tools: drawing the solution before you build it** in Innovative Solutions. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

□ The exact dot point

From the VCE Applied Computing Study Design (Unit 2, Outcome 1, Key Knowledge):

“design tools and techniques for representing solution designs, such as mock-ups, pseudocode, sitemaps and storyboards”

Jade’s team designs the FloatMate app

You met **Jade** earlier in this unit — the Year 11 Nippers volunteer who invented **FloatMate**, a wearable wristband that warns swimmers when they drift into a rip current. The wristband works. Now her team has to design the **companion phone app** that pairs with it: a screen for beach status, push alerts, settings, and a help guide.

Because Outcome 1 is a **collaborative** task, Jade can’t just build the app in her head — she has to *show* her teammates and her mentor what she means. That’s exactly what design tools are for: **representing the design so other people can see it, question it, and improve it before a single line of code is written.**

Detects drift, buzzes the wearer (built in KK01–KK04).

Beach status, alerts, settings, help — needs designing.

Jade + 3 students + a surf-club mentor. They need shared blueprints.

One design, four lenses

Here’s the FloatMate app as a stack of screens. Switch the **lens** to see how each design tool represents the *same* app in a completely different way. Notice: nothing about the app changes — only what the tool chooses to **show**.

Same content, four questions: Mock-up *what does it look like?* · Sitemap *how do the screens connect?* · Storyboard *what happens, in what order?* · Pseudocode *what logic runs underneath?*

Flip to lock in the one-line definitions

A picture of one screen's layout & look.

A visual draft of **a single screen** showing the position, size and style of every component (buttons, text, images, colours, fonts).

A tree of all pages and their links.

A hierarchical diagram of **every page/screen** and how they are organised and linked — the navigation map of the whole solution.

Frames in order showing a journey.

A series of frames in **time order** showing the path a user takes through screens for one task — this, then this, then this.

Plain-English steps of the logic.

Structured plain-English steps of an **algorithm's logic** — sequence, selection and iteration — without being tied to any one programming language.

Mock-up What the screen looks like

A **mock-up** is a visual draft of **one screen**. It shows where each component sits, how big it is, and how it's styled — the layout, fonts, colours, icons and labels. It deliberately says nothing about navigation or logic; it answers a single question: *"what will the user see?"*

A **low-fidelity** mock-up (a wireframe) is a quick boxes-and-lines sketch to test layout fast. A **high-fidelity** mock-up looks close to the finished product — real colours, real fonts. Early in design you sketch low-fi; later you polish to high-fi.

An **annotated** mock-up adds *callout labels* explaining each component — its data, its formatting, what happens when tapped. Annotations turn a pretty picture into a buildable specification the developer can follow. **Examiners reward annotation.**

Try it — tap a component to read its annotation

This is the FloatMate **Beach Status** screen. Each dashed region is an annotatable component. Tap one.

App title, centred. Font: Inter Bold 18pt, white on dark-teal (#1d2b45). Fixed at top on every screen for branding & orientation.

Largest element — draws the eye first. Status word pulled live from the band feed (*SAFE* green / *CAUTION* amber / *RIP* red). Beach name + timestamp above; plain-English sub-line below.

Secondary info, two label-value rows. Smaller, dimmed text so it doesn't compete with the status card. Values refresh every 30 s.

Full-width, accent fill, high contrast. On tap → sends a test buzz to the paired band and shows a confirmation toast. The only tappable control on this screen.

Sitemap How the screens connect

A **sitemap** zooms out. Instead of one screen, it shows **all** of them as a **hierarchy** — a tree with the home screen at the top and branches to each section and subpage. Lines mean “you can navigate from here to there.” It answers: “*how is the whole thing organised, and what links to what?*”

- Every page/screen in the solution
- The **hierarchy** — top-level vs sub-pages
- The **links** between pages (navigation paths)
- How deep the structure goes (how many taps to anything)
- What any single page *looks* like (that’s a mock-up)
- The *order* a user moves through for a task (that’s a storyboard)
- Any processing or logic (that’s pseudocode)

Storyboard What happens, in what order

A **storyboard** borrows its name from film. It’s a row of **frames in time order** showing the **journey** a user takes through screens to complete *one* task. Arrows between frames mean “next.” It answers: “*walk me through what the user actually does, step by step.*”

Below are the five frames of FloatMate’s most important journey — “**a rip is detected while Sam is swimming.**” They’ve been **shuffled**. Tap them into the right order. When the sequence is correct, the storyboard plays.

Pseudocode The logic underneath

The first three tools are about the *interface*. **Pseudocode** is the odd one out — it represents the **processing logic**: the algorithm. It’s written in structured plain English using the three control structures — **sequence**, **selection** and **iteration** — so any developer can read it and code it in *any* language. It is *not* real code and a computer can’t run it.

Here’s the logic behind FloatMate’s alert — the brains the mock-up button and storyboard are sitting on top of:

- **Sequence** — steps run top to bottom
- **Selection** — IF ... THEN ... ELSE ... END IF
- **Iteration** — WHILE ... END WHILE / FOR ... END FOR
- Keywords in CAPS, meaningful names, indentation, one action per line

It’s **language-independent** and readable by non-programmers on the team, so the design can be discussed and checked *before* committing to syntax. It captures the *logic* without drowning it in semicolons and brackets.

Which tool for which job?

This is where most marks live. The exam describes a need; you pick the right tool **and justify it**. Use the keyword in the question as your signal.

Drill — match the job to the tool

Pick the best design tool for each request from Jade’s team. Immediate feedback.

The architect analogy

Imagine designing a **house** instead of an app. The same four tools appear — and the analogy is worth quoting in an extended-response answer because it shows you understand *why each tool exists*.

The **interior render** of one room — what it looks like, finishes and all.

The **floor plan** — which rooms exist and which doors connect them.

The **walkthrough** — front door → hallway → kitchen, in the order a guest moves.

The **wiring logic** — IF the switch is flipped THEN the light turns on.

Command words → what to actually do

Give the tool, nothing more. 1 mark. “A storyboard.”

Say what the tool is *and* what it represents. “A sitemap is a hierarchical diagram showing all pages and how they link.”

Describe *plus* why/how it works in the context. Add the reason it suits the job.

Make the *difference* explicit in one sentence. “Whereas a mock-up shows one screen’s look, a storyboard shows the sequence across screens.”

Pick a tool and defend it against the scenario’s needs. This is where the architect analogy and “communicate to the team” lines pay off.

Actually produce the artefact — a labelled mock-up, a tree sitemap, ordered frames, or indented pseudocode. Use the conventions.

P-E-L writing trainer

High-mark responses follow **P-E-L — Point, Evidence/Example, Link**. Pick a prompt, write your answer, and watch the keyword tracker light up as you hit the ideas examiners reward.

Distinguish between a mock-up and a storyboard as design tools, using FloatMate as an example. (Aim: Point → Evidence → Link.)

VCE-style questions with model answers

Attempt each in your head or on paper first. Then reveal the **weak** answer (and *why* it loses marks), the **strong** answer, and the **marking guide**. Mark yourself — your score feeds the dial in the corner.

- **1** — correctly names **sitemap**.
- **1** — identifies structure/links across all pages, explicitly beyond what a mock-up (single-screen appearance) shows.
- **1** — mock-up correctly described (single screen / appearance / layout) with an example.
- **1** — storyboard correctly described (ordered sequence across screens) with an example.
- **1** — explicit statement of the difference (appearance of one vs sequence across many).
- a) **1** mock-up + **1** justification referencing visual/layout detail (annotation a bonus).
- b) **1** pseudocode + **1** justification referencing it representing decision/IF logic.
- No marks for naming a tool that can’t do the job (e.g. storyboard for logic).
- **1** — correct condition using both limits with AND.
- **1** — correct assignment of “CAUTION” / “SAFE” to status in the right branches.
- **1** — correct selection structure & conventions (IF/THEN/ELSE/END IF, indentation).

- **1** — clear statement of the purpose of representing a design (communicate/review before building).
- **3** — three different tools each correctly tied to *how* they support the design (1 each).
- **1** — explicit link to collaboration / team alignment / early feedback.
- **1** — coherent, well-structured explanation using the scenario (not generic).

Cheat sheet

- Visual draft of **one screen**
- Shows layout, size, fonts, colours, icons
- **Annotate** components for marks
- Low-fi sketch → high-fi polished
- Hierarchy of **all pages**
- Shows organisation + links (navigation)
- Non-linear map of options
- No look, no order, no logic
- Frames in **time order**
- Shows the user's journey/flow for one task
- Several screens, low detail, arrows = next
- Sequence, not structure
- Structured plain-English **logic**
- Sequence · selection · iteration
- Language-independent, not executable
- The only one for processing/decisions
- **Name** the tool → **say what it represents** → **justify** against the scenario's need (often: "so the team can review the design before building").
- Keyword cheat: *look*→mock-up · *link/structure*→sitemap · *order/flow*→storyboard · *logic/decision*→pseudocode.

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Common traps & misconceptions

Exam trap

TRAP 1 Confusing a storyboard with a mock-up **Fix:** mock-up = *one* screen in detail; storyboard = *several* screens in order. If the keyword is "sequence / flow / journey," it's a storyboard.

Exam trap

TRAP 2 Confusing a sitemap with a storyboard **Fix:** a sitemap is a non-linear *map of options* (a hierarchy). A storyboard is a linear *ordered path* through it. Map vs route.

Exam trap

TRAP 3 Treating pseudocode as "real code" **Fix:** pseudocode is structured plain English; it's *language-independent* and not executable. Don't write Python and call it pseudocode.

Exam trap

TRAP 4 Giving an un-annotated mock-up **Fix:** annotate. Label components, data, formatting and behaviour. Annotations are where the “design represents the solution” marks come from.

Exam trap

TRAP 5 Naming a tool but not justifying it **Fix:** “use a sitemap” earns little. “use a sitemap *because the team needs to agree on navigation structure across all five screens*” earns the marks.

Exam trap

TRAP 6 Forgetting the purpose of design tools **Fix:** they *represent the design so it can be communicated, reviewed and improved before development* — especially in a collaborative team. Say that.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. State

Jade’s team wants to **agree on how the FloatMate app’s five screens link together** before they start building. a) Name the most appropriate design tool. **(1)** b) State one thing this tool shows that a mock-up does not. **(1)**

✗ A weak answer

a) A storyboard. b) It shows the screens of the app. a) Wrong tool — a storyboard shows an ordered journey, not how screens link. b) Too vague and doesn’t contrast with a mock-up — no mark.

✓ A strong answer

a) A **sitemap**. b) It shows the **navigation structure/links between all five pages**, whereas a mock-up only shows the appearance of a single screen.

How the marks are awarded

- **1** — correctly names **sitemap**.
- **1** — identifies structure/links across all pages, explicitly beyond what a mock-up (single-screen appearance) shows.

Q2. Distinguish

Distinguish between a mock-up and a storyboard as design tools, using the FloatMate app as an example. **(3)**

✗ A weak answer

A mock-up is a drawing of the app. A storyboard is also a drawing of the app but it tells a story with the screens. No clear point of difference — both are called “a drawing.” No real FloatMate example. Misses the key contrast: one screen’s appearance vs a sequence across screens.

✓ A strong answer

A **mock-up** is a visual draft of a **single screen** showing its layout and appearance — e.g. the Beach Status screen’s header, status card and button. A **storyboard** is a series of frames in **time order** showing the user’s journey across **several** screens — e.g. the five frames of a rip being detected. **The difference:** a mock-up captures the appearance of one screen, whereas a storyboard captures the flow/sequence across multiple screens.

How the marks are awarded

- **1** — mock-up correctly described (single screen / appearance / layout) with an example.
- **1** — storyboard correctly described (ordered sequence across screens) with an example.
- **1** — explicit statement of the difference (appearance of one vs sequence across many).

Q3. Recommend

The team is collaborating on the app. For **each** of the following, recommend an appropriate design tool and justify your choice. a) Showing the developer the exact layout, colours and fonts of the alert screen. **(2)** b) Representing the decision the app makes about when to show a RIP alert. **(2)**

✗ A weak answer

a) A mock-up because it shows the screen. b) A storyboard to show the steps of the alert. a) Right tool, but “shows the screen” is a weak justification and ignores annotation. b) Wrong tool — a storyboard can’t represent branching decision logic; that needs pseudocode.

✓ A strong answer

a) An **annotated mock-up** — it visually shows the position, size, colour and font of each component on that one screen, and annotations let the developer build it exactly as designed. **(2)** b) **Pseudocode** — it expresses the selection logic (IF drift > limit AND in water THEN status “RIP”) in language-independent structured English the developer can translate into any language. **(2)**

How the marks are awarded

- a) **1** mock-up + **1** justification referencing visual/layout detail (annotation a bonus).
- b) **1** pseudocode + **1** justification referencing it representing decision/IF logic.
- No marks for naming a tool that can’t do the job (e.g. storyboard for logic).

Q4. Explain

Jade's team has three weeks to **design the FloatMate app collaboratively** before development begins. **Explain** how design tools support this design stage, referring to at least **three** different design tools and the **purpose** of representing a design. **(6)**

✗ A weak answer

Design tools help you plan the app. A mock-up is a picture of a screen. A sitemap shows the pages. A storyboard shows screens in order. So design tools are useful for the team. States what three tools are, but not how each helps the design/team. No real explanation of the purpose of representing a design. No link to collaboration, review, or building the right thing — caps in the middle band.

✓ A strong answer

Purpose: design tools **represent the design visually and logically so it can be communicated, reviewed and improved before any code is written** — essential for a collaborative team that must agree on one shared vision and catch problems while they're still cheap to fix. • A **sitemap** lets the team agree on the **navigation structure** across all five screens, so everyone builds to the same map. • **Annotated mock-ups** let them agree on the **look** of each screen and give the developer an exact, buildable specification. • A **storyboard** lets them check the **alert journey** (rip detected band buzzes phone shows RIP return to shore) makes sense as a sequence. • (Optional fourth) **Pseudocode** lets them agree the **alert logic** independent of language. **Link:** because the team is collaborating, these shared representations keep everyone align

How the marks are awarded

- **1** — clear statement of the purpose of representing a design (communicate/review before building).
- **3** — three different tools each correctly tied to *how* they support the design (1 each).
- **1** — explicit link to collaboration / team alignment / early feedback.
- **1** — coherent, well-structured explanation using the scenario (not generic).